

Chapter 2: Cost Elements & Breakeven Analysis

1. Cost Classification

A. Variable Costs

Definition: Costs that **vary** with production volume

Type	Full Form	Examples
Direct Material Cost	Cost of raw materials	Steel, plastic, fabric, components
Direct Labour Cost	Wages for production workers	Assembly line wages, machine operators
Direct Expenses	Purchase-related costs	Freight, insurance, carriage, import duty

B. Overhead Costs (Fixed Costs)

Definition: Costs that remain **constant** regardless of production volume

Type	Full Form	Examples
Administration Overhead	Management costs	Legal fees, audit fees, office rent, salaries
Selling Overhead	Marketing & sales costs	Advertising, commission, showroom expenses
Distribution Overhead	Delivery costs	Packing, transportation, warehouse rent

2. Basic Revenue and Cost Formulas

Total Sales Revenue (S or TR)

$$S = s \times Q$$

Variable Definitions: | Symbol | Full Form | Unit | Description | | S (or TR) | Total Sales/Total Revenue | Rs., \$ | Total income from sales | | s | Selling price per unit | Rs./unit, \$/unit | Price at which each unit is sold | | Q | Quantity | units | Number of units produced/sold |

When to Use: Calculating total sales income at any output level

Example: “Selling price = \$50/unit, Quantity sold = 100 units. Find total sales.”

$$S = 50 \times 100 = \$5,000$$

Total Cost (TC)

$$TC = TVC + TFC$$

$$TC = (v \times Q) + FC$$

Solution:

$$BEQ = \frac{500}{50 - 30} = \frac{500}{20} = \boxed{25 \text{ units}}$$

Verification:

At $Q = 25$:

Revenue = $50 \times 25 = \$1,250$

Total Cost = $(30 \times 25) + 500 = \$1,250$

Profit = $1,250 - 1,250 = \$0$ (Breakeven confirmed)

4. Breakeven Sales

Formula:

$$\text{Breakeven Sales} = \frac{FC \times s}{s - v}$$

OR

$$\text{Breakeven Sales} = BEQ \times s$$

Variable Definitions: | Symbol | Full Form | Unit | Description | |
Breakeven Sales | Breakeven Sales Revenue | Rs., \$ | Sales revenue at breakeven point |

When to Use: Finding sales revenue (not units) at breakeven

Example: Using previous example ($BEQ = 25$ units, $s = \$50$)

Breakeven Sales = $25 \times 50 = \$1,250$

OR

Breakeven Sales = $(500 \times 50) / (50 - 30) = 25,000 / 20 = \$1,250$

5. Contribution

Formula:

$$\text{Contribution per unit} = s - v$$

$$\text{Total Contribution} = Q(s - v)$$

Variable Definitions: | Symbol | Full Form | Description | |
Contribution Margin | Amount available to cover FC and profit |

When to Use: Building block for BEQ formula

Analyzing profitability per unit

Calculating P/V ratio

Example: “ $s = \$50$, $v = \$30$. Find contribution.”

Contribution per unit = $50 - 30 = \$20$

Interpretation: Each unit contributes \$20 towards fixed costs and profit

6. Margin of Safety

Formula:

$$\text{Margin of Safety} = \text{Actual Sales} - \text{Breakeven Sales}$$

$$\text{Margin of Safety \%} = \frac{\text{Actual Sales} - \text{BE Sales}}{\text{Actual Sales}} \times 100$$

Variable Definitions: | Symbol | Full Form | Description | |———|———|———|———| | Margin of Safety | Safety Buffer | Sales cushion above breakeven | | Actual Sales | Current Sales Revenue | Actual sales in Rs./\$ | | BE Sales | Breakeven Sales | Sales needed to breakeven |

When to Use: Measuring risk (how far sales can drop before loss)
Business safety assessment
Comparing investment alternatives

Example: “Actual sales = \$2,500, Breakeven sales = \$1,250. Find margin of safety.”

$$\text{Margin of Safety} = 2,500 - 1,250 = \$1,250$$

$$\text{Margin of Safety \%} = (1,250/2,500) \times 100 = 50\%$$

Interpretation: Sales can drop by 50% before business reaches breakeven

7. P/V Ratio (Profit/Volume Ratio)

Formulas:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$\text{P/V Ratio} = \frac{s - v}{s} \times 100$$

$$\text{P/V Ratio} = \frac{\Delta \text{Profit}}{\Delta \text{Sales}} \times 100$$

Variable Definitions: | Symbol | Full Form | Unit | Description | |———|———|———|———| | P/V Ratio | Profit-Volume Ratio | % | Contribution per rupee/dollar of sales | | Δ Profit | Change in Profit | Rs., \$ | Difference in profit levels | | Δ Sales | Change in Sales | Rs., \$ | Difference in sales levels |

When to Use: Finding contribution per rupee/dollar of sales
Calculating profit at different sales levels
Alternative method for breakeven calculations
Comparing profitability of products

Example: “ $s = \$50$, $v = \$30$. Find P/V ratio.”

$$\text{P/V Ratio} = [(50 - 30)/50] \times 100 = (20/50) \times 100 = 40\%$$

Interpretation: Every \$1 of sales contributes \$0.40 towards fixed costs and profit

Related P/V Formulas:

Profit using P/V Ratio:

$$\text{Profit} = (\text{Sales} \times \text{P/V Ratio}) - FC$$

Required Sales for Target Profit:

$$\text{Required Sales} = \frac{FC + \text{Desired Profit}}{\text{P/V Ratio}}$$

Example: “ $FC = \$500$, $P/V \text{ Ratio} = 40\%$, $\text{Desired profit} = \300 . Find required sales.”

$$\text{Required Sales} = (500 + 300) / 0.40 = 800 / 0.40 = \$2,000$$

8. COMPLETE EXAMPLE: Breakeven Table Analysis

Problem Statement:

“A firm produces units with the following cost structure. Selling price is \$50 per unit. Find: (a) Total cost and sales at each output level, (b) Profit at each level, (c) Breakeven point, (d) Marginal Revenue, (e) Marginal Cost.”

Given Data:

Output (Q)	TVC ()	TFC()
0	0	500
10	300	500
20	600	500
25	750	500
30	900	500
40	1200	500
50	1500	500

Where: - Q = Quantity/Output (units) - TVC = Total Variable Cost () - **TFC** = TotalFixedCost()
- s = Selling price = \$50/unit

Solution:

Step 1: Calculate Total Cost (TC)

$$TC = TVC + TFC$$

Q	TVC	TFC	TC = TVC + TFC
0	0	500	500
10	300	500	800
20	600	500	1100
25	750	500	1250
30	900	500	1400
40	1200	500	1700

Q	TVC	TFC	TC = TVC + TFC
50	1500	500	2000

Step 2: Calculate Total Sales (S)

$$S = s \times Q = 50 \times Q$$

Q	S = 50 × Q
0	0
10	500
20	1000
25	1250
30	1500
40	2000
50	2500

Step 3: Calculate Profit

$$\text{Profit} = S - TC$$

Q	S	TC	Profit = S - TC	Status
0	0	500	-500	Loss
10	500	800	-300	Loss
20	1000	1100	-100	Loss
25	1250	1250	0	BREAKEVEN
30	1500	1400	+100	Profit
40	2000	1700	+300	Profit
50	2500	2000	+500	Profit

Breakeven Point: Q = 25 units (Profit = 0)

Step 4: Calculate Variable Cost per Unit (v)

$$v = \frac{TVC}{Q} = \frac{1500}{50} = \$30 \text{ per unit}$$

Verification: At Q = 10: v = 300/10 = \$30 (Constant)

Step 5: Verify BEQ using Formula

$$BEQ = \frac{FC}{s - v} = \frac{500}{50 - 30} = \frac{500}{20} = \boxed{25 \text{ units}} \checkmark$$

Matches table result!

Step 6: Calculate Marginal Revenue (MR)

$$MR = \frac{\Delta S}{\Delta Q} = \frac{\text{Change in Sales}}{\text{Change in Quantity}}$$

For **linear pricing** (constant price):

$$MR = s = \$50 \text{ per unit}$$

Why? Each additional unit sells for \$50

Step 7: Calculate Marginal Cost (MC)

$$MC = \frac{\Delta TC}{\Delta Q}$$

Q → Q	ΔTC = TC - TC	ΔQ = Q - Q	MC = ΔTC/ΔQ
0 → 10	800 - 500 = 300	10	30
10 → 20	1100 - 800 = 300	10	30
20 → 25	1250 - 1100 = 150	5	30
25 → 30	1400 - 1250 = 150	5	30
30 → 40	1700 - 1400 = 300	10	30
40 → 50	2000 - 1700 = 300	10	30

MC = \$30 per unit (constant, equals v)

Why constant? Because variable cost per unit is constant

Step 8: Calculate P/V Ratio

$$\text{P/V Ratio} = \frac{s - v}{s} \times 100 = \frac{50 - 30}{50} \times 100 = \frac{20}{50} \times 100 = \boxed{40\%}$$

Interpretation: Each \$1 of sales contributes \$0.40 to fixed costs and profit

Step 9: Calculate Margin of Safety (at Q = 50)

$$\text{Actual Sales} = \$2500$$

$$\text{BE Sales} = \$1250$$

$$\text{Margin of Safety} = 2500 - 1250 = \boxed{\$1250}$$

$$\text{Margin of Safety \%} = \frac{1250}{2500} \times 100 = \boxed{50\%}$$

Interpretation: Sales can drop 50% before reaching breakeven

Summary of Results:

Metric	Value	Formula Used
Breakeven Quantity	25 units	$FC/(s-v)$
Breakeven Sales	\$1,250	$BEQ \times s$
Variable Cost per unit	\$30	TVC/Q
Contribution per unit	\$20	$s - v$
Marginal Revenue	\$50/unit	s (constant price)
Marginal Cost	\$30/unit	$\Delta TC/\Delta Q$
P/V Ratio	40%	$(s-v)/s \times 100$
Margin of Safety (Q=50)	\$1,250 (50%)	Actual Sales - BE Sales

Key Takeaways:

1. **Below BEQ ($Q < 25$):** Loss
 2. **At BEQ ($Q = 25$):** Zero profit (breakeven)
 3. **Above BEQ ($Q > 25$):** Profit increases by \$20 per unit (contribution)
 4. **MR > MC:** \$50 > \$30, so producing more units increases profit
 5. **P/V Ratio = 40%:** Every additional \$1 sales adds \$0.40 profit (after breakeven)
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